

Ta₄C₃T_x MXene — Single-/Few-Layer Powder

Research-grade two-dimensional transition-metal carbide powder

Ta₄C₃T_x MXene is supplied as a delaminated single-/few-layer powder derived from Ta₄AlC₃. The material has a representative layer count of ≤5. The grade is supplied for electrochemical, energy-storage, conductive-composite, interface and broader two-dimensional-material research.

KEY FEATURES

- Delaminated single-/few-layer powder derived from Ta₄AlC₃
- Nominal lateral size: 200–1000 nm
- Typical conductivity of 10–500 S/cm
- Surface terminations: –O, –OH, –F, –Cl

POTENTIAL RESEARCH APPLICATIONS

- Electrochemical and energy-storage material research
- Conductive-composite research
- Interface and two-dimensional-material research

STORAGE & HANDLING

Store dry in a tightly closed container. Vacuum or inert-atmosphere storage is recommended. Minimize air and moisture exposure.

QUALITY / DOCUMENTATION

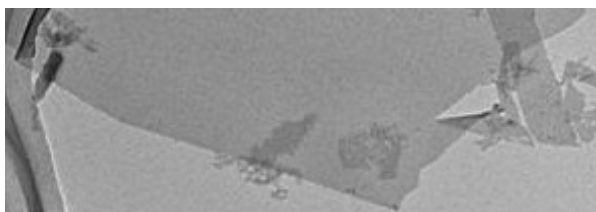
Surface termination distributions vary with synthesis and lot. Quantitative surface composition can be discussed when required.

TECHNICAL SPECIFICATIONS

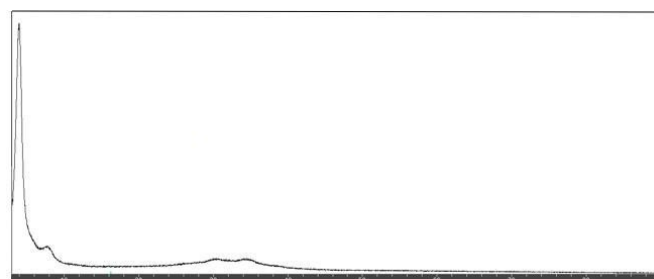
PROPERTY	TYPICAL / REPRESENTATIVE VALUE
Material	Ta ₄ C ₃ T _x MXene
Precursor	Ta ₄ AlC ₃
Product form	Single-/Few-Layer Powder
Preparation	HF etching followed by intercalation/delamination
Layer structure / count	≤5 layers
Nominal lateral size	200–1000 nm
Typical conductivity	10–500 S/cm
Surface terminations	–O, –OH, –F, –Cl
Recommended storage	Dry; vacuum or inert atmosphere

Conductivity values are representative technical values. Measurement results can vary with sample preparation, packing, film preparation and measurement geometry.

REPRESENTATIVE CHARACTERIZATION



Representative microscopy; not lot-specific.



XRD. Representative characterization; not lot-specific.
Non-data annotation removed; diffraction trace and axes otherwise unchanged.

WINIGEN MATERIALS

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Representative technical values. Properties can vary with synthesis and lot. Lot-specific documentation and additional analytical details are available subject to review.